

MB 585: Supply Chain Analytics

Course Code: MB 585

Course Name: Supply Chain Analytics

L-T-P-C: 2-0-0-2

Prerequisites: Supply Chain Management.

Intended for: MBA DS&AI

Mutual Exclusion: NA

Approval: 59th BoA (MBA DS&AI curriculum); course proposal approved in the 60th BoA

Preamble

In an increasingly dynamic and data-driven world, supply chains are evolving into complex, intelligent networks requiring robust analytical capabilities. This course on Supply Chain Analytics introduces students to the transformative role of analytics, AI, and digital technologies in modern supply chain management. Through a comprehensive curriculum that blends foundational concepts with advanced tools—ranging from predictive modeling and network optimization to machine learning and blockchain applications—the course empowers students to make data-informed, resilient, and strategic decisions across the supply chain spectrum. Designed for future leaders in the data science and AI domains, this course fosters a deep understanding of how analytics can drive efficiency, agility, and innovation in supply chain operations.

Course Modules

- **Introduction to Supply Chain Analytics:** What and why of Supply Chain Analytics, Competitive Analysis, and Benchmarking; Types of supply chain Analytics, Digital Transformation of the supply chain; Descriptive, predictive, and prescriptive analytics for supply chain, optimization and simulation for supply chain analytics. (6)
- **Supply Chain Models and Decisions:** Model for the application of Supply Chain Analytics; Perspectives of SC analysts and users on SC tools; analytical questions and the appropriate Supply Chain tools; Uncertainties in supply chains, Order Management, and Demand Fulfillment: Uncertainty Modelling Versus Demand Forecasting, Additive and Multiplicative demand models, Analytical Safety Stock Models; Modelling Flexibility in Supply Chain, Trends, Challenges; Decision-making across end-to end supply chain networks. (4)
- **Supply Chain Risk Analytics:** Supply Chain Disruptions, Supply Chain Risks and their characteristics; Supply Chain Risk Analysis and Analytics: Important Elements of Supply Chain Risk Analysis, Risk Acceptance, Reduction and Spreading, Supply Chain Risk Identification and Assessment. Supply chain resilience and its analytics. (4)
- **Network Optimization:** Supply chain analytics to facility location planning and network design, optimal warehouse locations and factories, Optimal locations of several warehouses and factories, stochastic supply network design. (4)
- **AI for Supply Chain Analytics:** Digital twin-based Supply chain and its analytics, Machine Learning (ML) for Supply Chain Planning, supplier selection, and warehouse management, ANNs in Inventory, Machine Learning-based Newsvendor models, Autonomous Vehicles for Logistics and Shipping, Natural language processing (NLP) for procurement. (6)

- **Blockchains for Supply Chain:** Blockchains in Supply Chain Management, Blockchain technology for supply chain visibility, predictability, and controllability, Digital tokens in supply chain transactions, Challenges in Blockchain Supply Chains, Application of Blockchain in Transportation and Logistics, Blockchains in Supply Chain Traceability, Machine-as-a-Service leasing model. (4)

Text Books

Instructor to provide reference material and cases.

Reference Books

1. Dmitry Ivanov, *Introduction to Supply Chain Analytics with Examples in AnyLogic and anyLogistix Software*, Springer Nature Switzerland AG 2024.
2. Iris Heckmann, *Towards Supply Chain Risk Analytics: Fundamentals, Simulation, Optimization*, Springer Gabler, Springer Fachmedien Wiesbaden 2016.
3. Işık Biçer, *Supply Chain Analytics: An Uncertainty Modeling Approach*, Springer Nature Switzerland AG 2023.